Bihar Engineering University, Patna

B.Tech. 8th Semester Examination, 2024

Course: B.Tech. *Time: 03 Hours*
Code: 100817 *Subject: Cloud Computing* *Full Marks: 70*

Instructions:-
(i) The marks are indicated in the right-hand margin.
*(ii) There are **NINE** questions in this paper.*
*(iii) Attempt **FIVE** questions in all.*
*(iv) Question No. **1** is compulsory.*

Q.1 Choose the correct option / write the answer of the following (any seven): **|2 x 7 = 14|**

(a) Which of the following is a characteristic of Cloud Computing?
- (i) On-demand self-service
- (ii) Limited network access
- (iii) Fixed storage capacity
- (iv) High maintenance costs

(b) Which cloud service model provides hardware resources over the internet?
- (i) SaaS
- (ii) PaaS
- (iii) IaaS
- (iv) DaaS

(c) What is elasticity in cloud computing?
- (i) The ability to run multiple applications simultaneously
- (ii) The ability to adapt to changing workload demands
- (iii) The ability to connect to various cloud providers
- (iv) The ability to recover from failures

(d) What is the primary function of a hypervisor in cloud computing?
- (i) To manage data storage
- (ii) To create and manage virtual machines
- (iii) To enhance network security
- (iv) To provide user interfaces

(e) Which of the following best describes 'scalability' in cloud computing?
- (i) Ability to downgrade services
- (ii) Ability to handle increased workloads
- (iii) Ability to provide continuous uptime
- (iv) Ability to ensure data security

(f) Which of the following is NOT a benefit of cloud computing?
- (i) Cost efficiency
- (ii) Increased security
- (iii) Flexibility
- (iv) On-premises hardware dependency

(g) Which of the following is a major concern in cloud computing?
- (i) Scalability
- (ii) Security
- (iii) Cost
- (iv) Flexibility

(h) What is the primary advantage of using cloud storage?
- (i) High cost
- (ii) Limited accessibility
- (iii) Scalability and flexibility
- (iv) Manual maintenance

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(i) What is 'serverless computing'?
   (i) Running applications without a server
   (ii) Outsourcing server management to a cloud provider
   (iii) Using servers without virtualization
   (iv) Automatically scaling applications based on demand

(j) What is the main purpose of a Content Delivery Network (CDN) in cloud computing?
   (i) To improve database performance
   (ii) To enhance application development
   (iii) To distribute content closer to end-users
   (iv) To secure cloud resources

Q.2 (a) What are the different types of Cloud Computing deployment models? What are the main characteristics of Cloud Computing? **[7]**

(b) What are the key factors to consider when choosing between different cloud service providers (e.g., AWS, Azure, Google Cloud)? **[7]**

Q.3 (a) Explain the internal workings of virtual machine monitors (VMMs). How do they manage hardware resources and ensure isolation between virtual machines? Explain in detail. **[7]**

(b) Characterize the following three cloud computing models: (a) What is an IaaS? Give one example system. (b) What is a PaaS? Give one example system. (c) What is a SaaS? Give one example system. **[7]**

Q.4 (a) In the process of VM (Virtual Machine) migration to a new host within the same LAN (Local area network), how will the existing communication's messages over the network be re-routed to the new host? Explain in detail. **[7]**

(b) Explain the different cloud resource provisioning plans with their advantages and disadvantages. **[7]**

Q.5 (a) Describe various scheduling and load balancing techniques used to improve Quality of Service (QoS) in cloud computing. **[5]**

(b) Briefly explain the following terms associated with network threats or security defense in a distributed computing system: (a) Denial of service (DoS) (b) Trojan horse (c) Network worm (d) Service spoofing (e) Authorization (f) Authentication (g) Data integrity (h) Confidentiality **[9]**

Q.6 (a) Discuss the role of Service Level Agreements (SLAs) in resource management and their impact on resource allocation and performance guarantees. **[7]**

(b) Explain the key principles of application-level security in cloud computing. **[7]**

Q.7 (a) What are the security implications of using multi-tenancy in cloud computing, and how can these risks be mitigated? **[7]**

(b) Explain the concept of Fog Computing and its role in reducing latency and bandwidth usage in IoT applications. How does it differ from traditional cloud computing? **[7]**

Q.8 (a) What are the key environmental benefits of Green Cloud Computing? What strategies can be employed to achieve energy efficiency in data centres? **[7]**

(b) Explain the concept of hypervisors in virtualization. How do Type 1 and Type 2 hypervisors differ, and what are their respective advantages and disadvantages? **[7]**

Q.9 Write Short notes on ***any two*** of the following:- **[7 x 2=14]**
(a) Types of Cloud Computing?
(b) Explain about Amazon EC2?
(c) Describe Hypervisor and Xen architecture?

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